

“HINDHIVAC”
MINI PIRANI STABILIZED GAUGE
MODEL – A6STM-D

OPERATION
AND
MAINTENANCE MANUAL



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CONTENTS

SL. NO.	TITLES	PAGE NO.
1.0	INTRODUCTION	3
2.0	GENERAL DESCRIPTION	3
3.0	SPECIFICATIONS	4
4.0	UNPACKING AND PRE-OPERATIONAL CHECK	4
5.0	INSTALLATION	4
6.0	CIRCUIT DESCRIPTION	5
6.1	BRIDGE VOLTAGE	5
7.0	PRINCIPLE OF OPERATION	6
7.1	OUTPUT FOR VACUUM CONTROLLER	7
8.0	OPERATION PROCEDURES	7
8.1	LEAK DETECTION	8
9.0	MAINTENANCE	8
9.1	RECONDITIONING OF THE FILAMENT	8
9.2	REPLACEMENT OF GAUGE HEAD TUBE	9
9.3	GAUGE CALIBRATION	9
10.0	TROUBLE SHOOTING	11

1.0 INTRODUCTION

The **HINDHIVAC** mini Pirani stabilized gauge **Model: A6STM- D** is a pressure control instrument designed with the thermal conductivity type gauge head (Sensor). This provides all necessary bridge circuits and signal conditioned analog outputs. This instrument can accept range from 0.5m.bar to 0.001 m.bar using a direct reading meter. This instrument works on the principle of change in resistance of material with a change in temperature. This can be used to measure vacuum from 0.5m.bar to 0.001m.bar anywhere in the vacuum system with a suitable adopter. This gauge is very much essential in the Pirani controller to control vacuum (0.5m.bar to 0.001m.bar). Two gauge heads can be directly connected to this gauge to read fore vacuum and roughing vacuum of a vacuum system.

2.0 GENERAL DESCRIPTION

The mini Pirani gauge **Model: A6STM - D** is a modular front panel construction and forms $\frac{1}{4}$ size of a standard 19" rack. This can be used as a bench standing type or as a panel mounting type. By using side clamps for both side of Pirani Gauge.

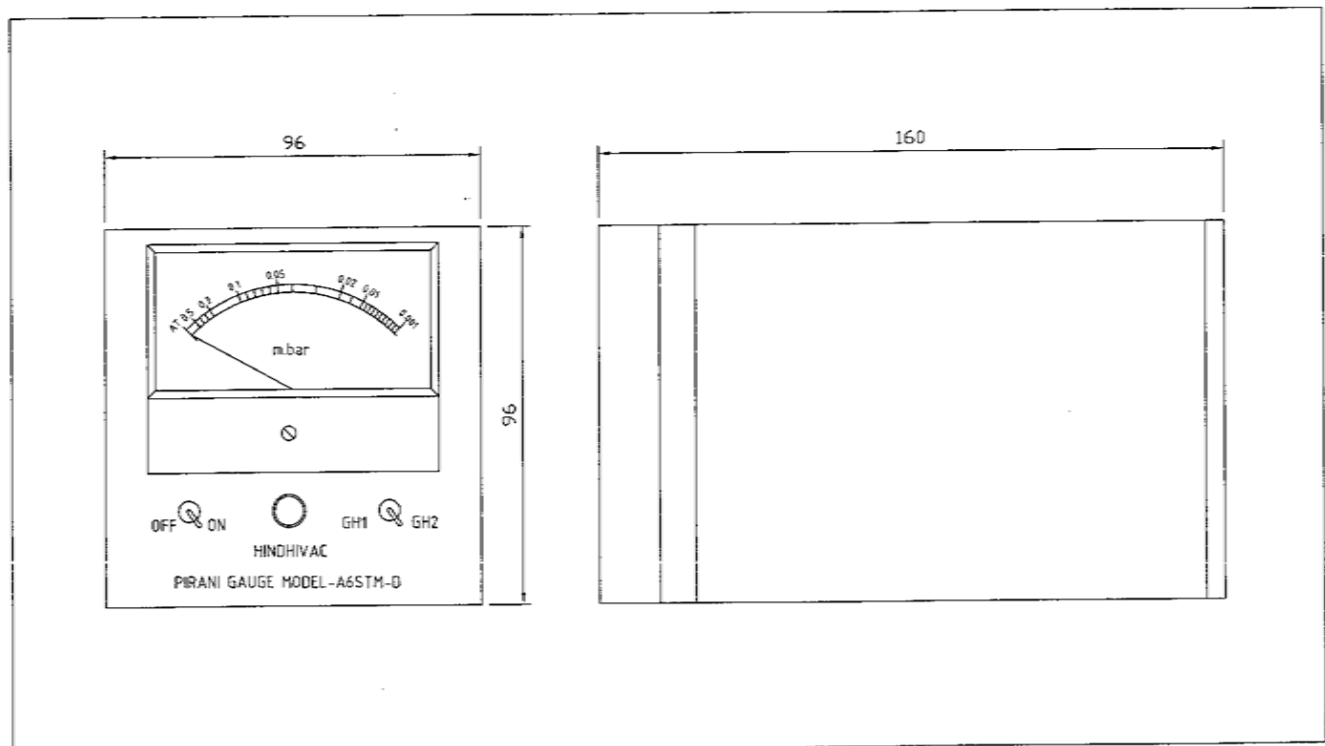


Fig.1 Diagram of Pirani Gauge Outline

This instrument is constructed with metal enclosures for top & bottom side of instrument. This facilitates, easy accessibility of all components mounted inside and the gauge makes the service more convenient. This gauge measures the pressure in the range of 0.5m.bar to 0.001m.bar using a direct reading meter. It operates with one or two **HINDHIVAC** Pirani Gauge Heads, which facilitate the measurement of the fore vacuum and roughing vacuum in any vacuum system.

The gauge is calibrated for dry air but will be found accurate for most purposes under normal working conditions, where small quantities of other gases or vapours are present

3.0 SPECIFICATIONS

ELECTRICITY SUPPLY	:	230V AC (+ OR – 10%)
BRIDGE OUTPUT VOLTAGE	:	2V D.C. (+ OR – 0.5%)
PRESSURE RANGE	:	0.5 m.bar to 0.001 m.bar
NO. OF GAUGE HEADS	:	TWO HINDHIVAC MODEL PR-3
FUSE	:	100 m. A. SIZE: 5 X 20 MM
GH1, GH2	:	INPUT TO HINDHIVAC PIRANI CONTROLLER MODEL PRGC-1 FOR PRESSURE CONTROL.
SIZE IN MM	:	96HX96WX160D. (APPROX.)

4.0 UNPACKING AND PRE-OPERATIONAL CHECK

The control panel and gauge heads are carefully checked and packed before despatch. Prior to using the instrument carefully inspect the unit for any visual signs of transportation damages.

If any damage is observed, please report to **HINDHIVAC** and to the insurance to file a proper claim: Please send copy of change claim to **HINDHIVAC** Bangalore in order to expedite replacements or repair of the instrument.

If the unit fails to function properly contact our local office or please write to **HINDHIVAC**, Bangalore.

5.0 INSTALLATION

- Check that the supply voltage agrees with operating voltage of instrument.
- Check that the mains ON switch (SW1) is OFF in position.
- Check that the meter indicates zero if not adjust by the mechanical zero adjusting screw on the meter face.

- Plug the gauge heads 1 & 2 respectively into their sockets firmly at the rear panel.
- Connect the gauge head to the vacuum system. See that the 'O' ring is properly placed in the male joint. Tighten the locknut with the hand properly, so that the 'O' ring provides a leak tight joint.
- Switch on the control unit by switching on the toggle switch (S1). Now the neon lamp should glow indicating readiness of the power supply to gauge head.

6.0 CIRCUIT DESCRIPTION

The 230V AC ($\pm 10\%$) mains supply is given to the primary part of the transformer (T) through a fuse (F) and mains switch (SW1) located on the front panel. The mains indication lamp (PL) is connected across the primary part of the transformer (T) and it glows when the instrument is switched on under normal working condition.

The output from the secondary part of the transformer is rectified by diodes D01 to D04 and smoothened by capacitor C01. The rectified voltage is fed to an integrated circuit voltage stabilizer, the output of which can be adjusted to +2V by adjustment of a preset (PO1). A capacitor CO2 is connected across the output of the regulator to reduce ripples. Resistors RO2 and RO3 form two arms of a bridge network and the other two arms are formed by the Pirani Gauge Head.

Potentiometer P2 is used for adjusting the initial setting of zero at atmospheric pressure and potentiometer P3 is provided for calibration of the gauge head P2 and P3: They are located inside the gauge heads.

6.1 BRIDGE VOLTAGE

The bridge voltage is set at the factory and should require no further adjustment. However, before a control unit is used for the first time, check the voltage as described below with the gauge heads connected.

- a. Remove the top plate of the chassis.
- b. Connect the gauge to the mains voltage supply.
- c. Allow 10 minutes of warm up.
- d. Measure the output voltage of the regulator on the printed card across RO2 and RO3.
- e. Use a D.C. voltmeter with the range of 0 - 2.5V DC (Avometer 8 x is suitable).

The meter should read 2V DC. If the voltage is not correct adjust (PO1) until it reads 2V DC.

7.0 PRINCIPLE OF OPERATION

Change of pressure in the vacuum system brings about a rise or fall in number of gas molecules present and hence a rise or fall in thermal conductivity of the gas. Thus the heat loss of the constant voltage electrically heated filament in the system varies with the pressure.

The Pirani gauge head filament has high temperature co-efficient of resistance. So a slight change in the system pressure brings about useful change in the filament resistance resulting in an out of balance current which can be read as pressure on the meter. The Pirani gauge head indicates the total pressure of combined gases and condensable vapours in the system as the heated wire can lose heat (both gas and vapour molecules).

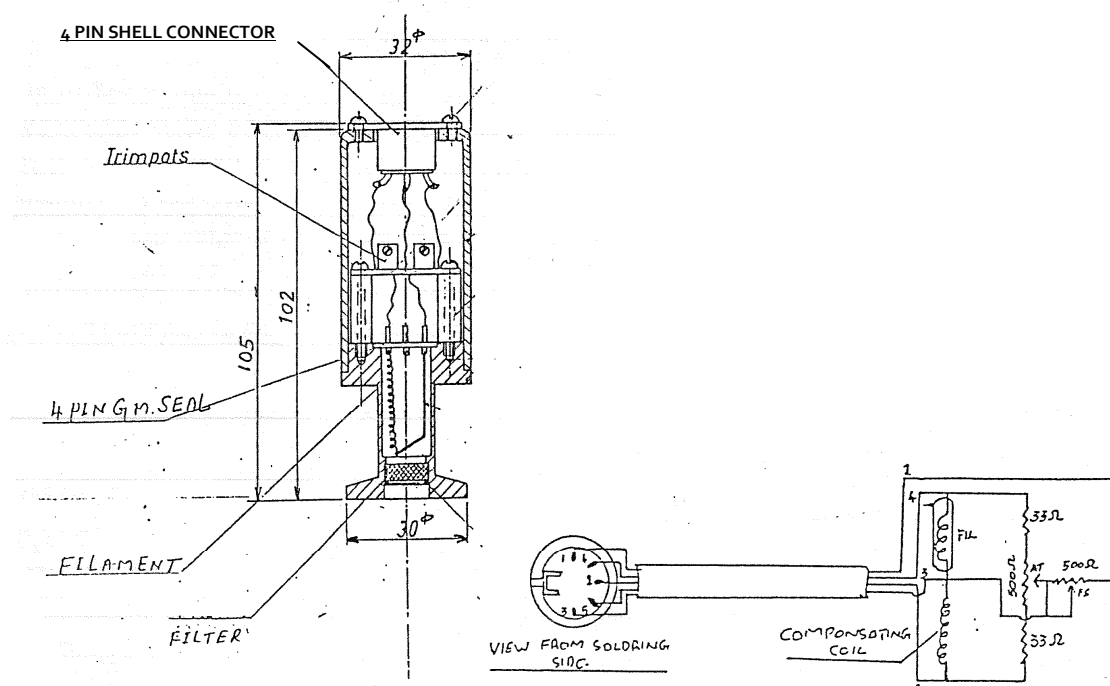


Fig.2 Diagram of Pirani Gauge Head

7.1 OUTPUT FOR VACUUM CONTROLLER

Two electrical signals which are proportional to the meter reading are provided on the back panel corresponding to GH1 and GH2 and marked as GH1 COM GH2. Any one of these signals can be connected to the **HINDHIVAC** Pirani controller Model PRGC-1, which works in the range of 0.5 m.bar to 0.001 m.bar to actuate any external electrical gadgets depend upon the preset vacuum.

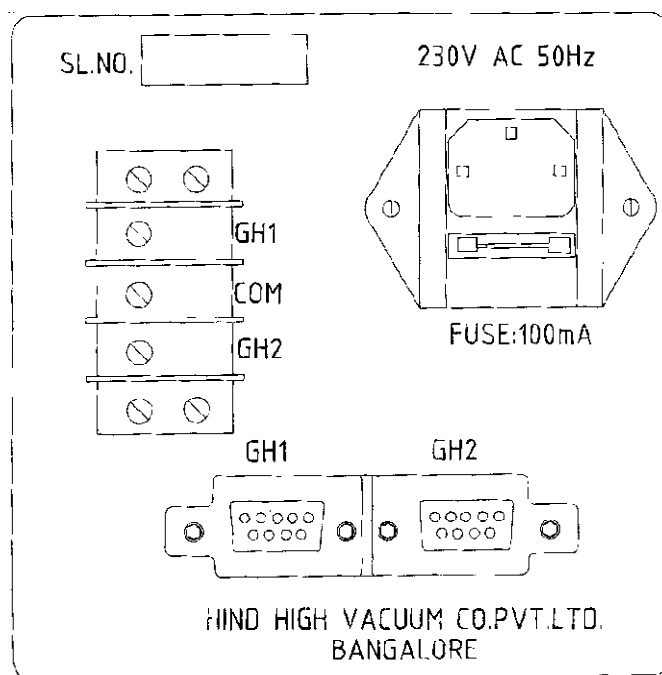


Fig.3 Diagram of Pirani Gauge Back Panel

8.0 OPERATION PROCEDURES

- Connect the 2 gauge heads at the rear panel.
- Switch ON the instrument by operating the mains ON switch.
- Keep SW2 in GH1 position.
- The meter should indicate 'AT' on the meter scale. If not, adjust (PO2) the thick wire potentiometer inside the gauge head until it reads 'AT'.
- Keep SW2 in G. Head-2 position. Similarly adjust the meter to read 'AT' by using a corresponding potentiometer. (Inside gauge head 2). Now the instrument is ready for vacuum measurement. Evacuate the system and observe the reading on meter scale.

8.1 LEAK DETECTION

The Mini Pirani Gauge can be used to locate coarse leaks such as leaks through 'O' ring seals, door flanges, windows, leak in shaft seals or valve diaphragm, hydrogen is ideally suitable as a probe gas in leak detection; Due to high rate of diffusion through the leak, the Pirani gauge senses almost immediately. Moreover the thermal conductivity of hydrogen at low pressure is almost twice that of dry air so that a Pirani gauge calibrated for dry air is twice as sensitive to hydrogen at the same absolute pressure.

The leak detection is to be carried on as follows:

Fit a hydrogen cylinder with a pressure reducing valve. An air gun nozzle or a piece of glass tubing of 1/8" diameter at one end and a small hole at the other end is recommended as a probe to direct a fine jet of hydrogen to the suspected leak area. Begin at the highest point in a vacuum system and work slowly and systematically downwards, observing the Pirani gauge for fluctuations and sealing each leak as it is found. In equipments with two or more gauge heads, select the gauge head nearest to the path of the gas between the probe area and the pumps for observation. Very small leaks require specialized leak detection apparatus.

PRECAUTION



Hydrogen is an explosive gas, necessary precautions should be taken while handling it. Alternatively, other safer gases can be used with reduced sensitivity which depends upon their thermal conductivity.

9.0 MAINTENANCE

Servicing of the gauge apart from the adjustments described should be undertaken only by an Electronic engineer familiar with ICs and Transistor circuits.

For voltage and current measurements use an AC/DC multi range meter having a sensitivity of atleast 20,000 ohms per volt.

9.1 RECONDITIONING OF THE FILAMENT

The gauge head behaves erratically it is filled with any contaminants, the gauge head has to be flushed out with suitable organic solvents such as

Petroleum, Ether, Benzene, Carbon tetrachloride or Acetone and then thoroughly dried.

Apply 10V AC or DC to pin No.1 and 4 of the gauge head plug for two minutes. This flashes the filament and volatilizes the deposits over the filament.

9.2 REPLACEMENT OF GAUGE HEAD TUBE

To replace the gauge head tube, remove the screws of the gauge head and take out the top cover. Now the metal body, compensating coil and potentiometers (PO2, PO3) are accessible. Desolder the leads of the metal gauge head from the cable and potentiometers (PO2, PO3) shorting point, introduce the new gauge head tube and reverse the procedures.

9.3 GAUGE CALIBRATION

When a new gauge head tube is replaced, it is necessary to calibrate the unit. The gauge is to be calibrated for dry air against a McLeod gauge with all the necessary traps etc., For accurate calibration, the unit has to be sent to the factory. However, for a rough calibration, the following procedure is to be adopted.

1. Follow procedure of operation procedures a,b,c,d and e.
2. Evacuate the gauge head to pressure below 0.001 m.bar.
3. Select SW2 switch to G.Head-1 position and adjust potentiometer PO2 (FS) (marked on the gauge head). Until the meter reads 0.001m.bar on the scale.
4. Select SW2 switch to G.Head-2 position and adjust potentiometer PO2 (FS) (marked on the gauge head) until the meter reads 0.001 m.bar on the scale.

PRECAUTION	The filament must be flashed only when the pressure in the gauge head is below 1 micron.
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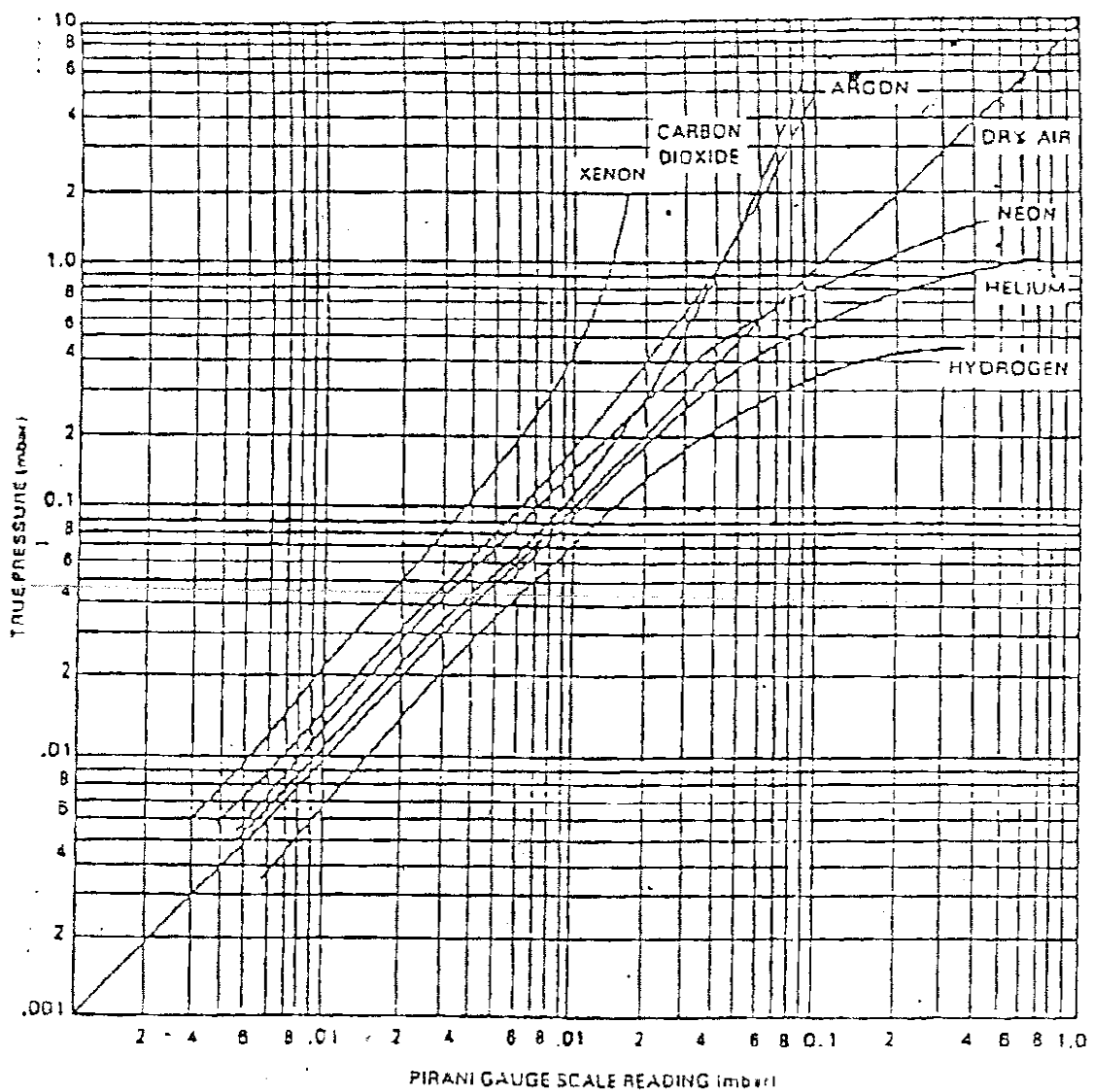


Fig.4 Calibration Chart

10.0 TROUBLE SHOOTING

SL. NO.	TROUBLE	CAUSE	REMEDY
1	Mains indicator lamp not glowing	Electricity supply has failed. Lamp PL has failed	Check supply voltage fuse-F power chord mains plug, switches SW1.
2	Bridge voltage is too high	IC is not working properly.	Check & replace
3	Bridge voltage is too low	Do1, DO2, DO3 & DO4 are faulty	Check & replace
4	Meter reads F.S.D. connection	Open gauge head or replace filament	Repair the gauge head
5	Pressure reading erratic	Gauge socket contacts are loose or dirty	Tighten or clean.
6	Calibration suspect	Bridge voltage set incorrectly	Carry out bridge voltage adjustment
7	No meter reading	Pressure at Atm. Or gauge head not plugged in or 500 ohms (F.S) potentiometer opened.	Check & replace

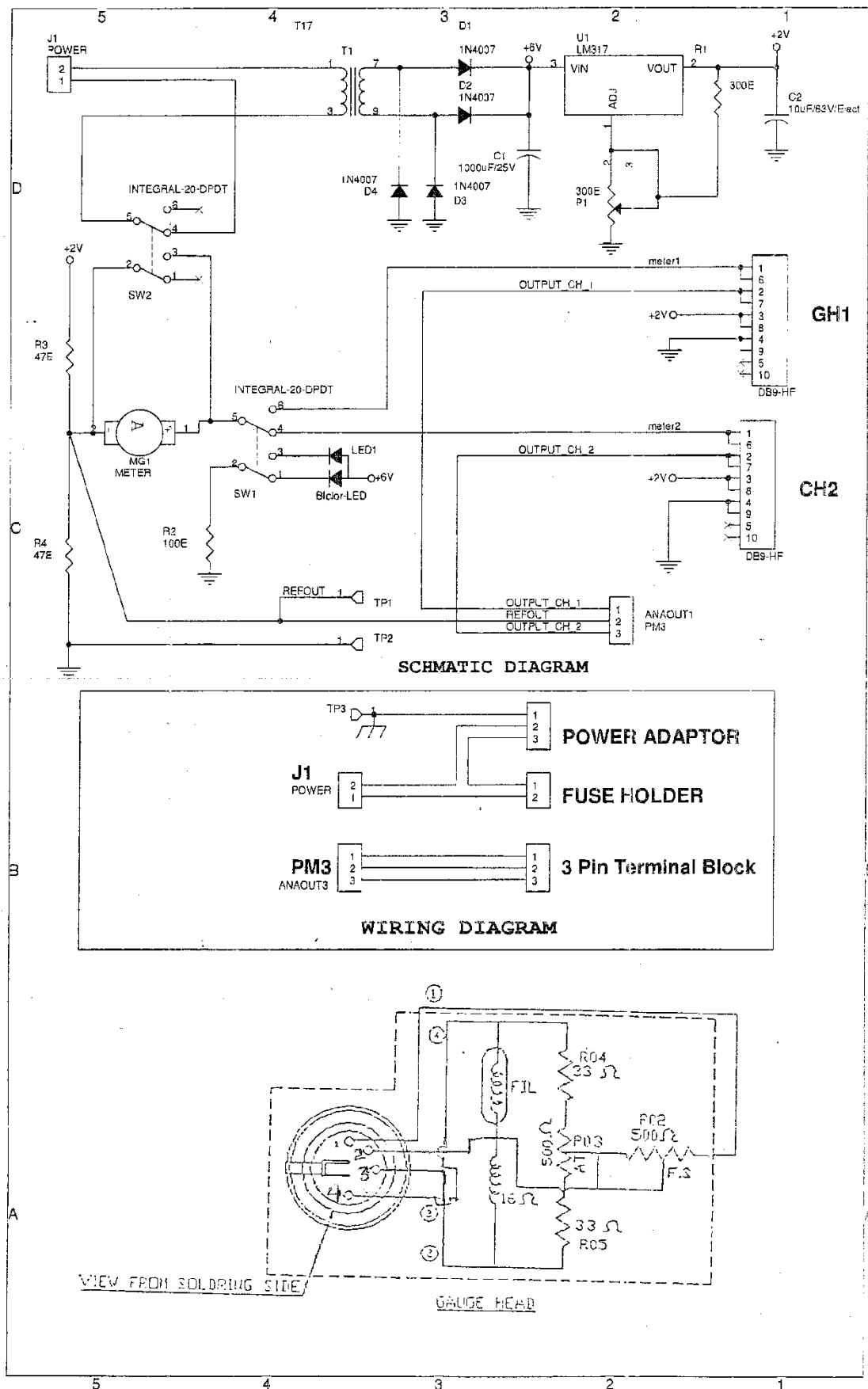


Fig.5 Circuit Diagram of Pirani Gauge Model: A6STM-D